

Lineaire Algebra H1–4: Overzicht

Studiegids uit de HC-aantekeningen (dictaat-nummers ter referentie)

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How the four weeks fit together (the big picture)

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Study guide built **only from the lecture notes (HC aantekeningen, weeks 1–4)**. The dictaat (Beukers, *Lineaire Algebra*, 2026 edition) is used purely as a reference: every definition/theorem below is tagged with its dictaat number so you can look it up.

Week	Notes cover	Dictaat sections	File
1	Vectors, lines, planes, inner product, plane equation	H1 (1.1–1.4), H2 (2.1–2.3, 2.5)	week1
2	Matrices, matrix product, Gauss–Jordan, pivots	§3.1, §3.2	week2
3	$n \times n$ systems, inverse matrix, subspaces, span, S_{hom} / S_{inhom}	§3.3, §3.4, §3.5	week3
4	Linear combination, (in)dependence, relations, rank, dimension, basis	§4.1, §4.2	week4

Not covered by the notes (and therefore not taught here): §2.4 worked examples, §2.6, §3.6, §3.7, §4.3 (rang van een matrix), §4.4 (somruimte). They are in the dictaat if you need them.

How the four weeks fit together (the big picture)

- Week 1** gives you the objects: vectors in $\mathbb{R}^n$, and two ways to describe lines/planes (parametrisation vs. equation). The inner product is the bridge between the two.
- Week 2** turns “solve a linear system” into a mechanical algorithm on a matrix (Gauss–Jordan). The output tells you: no solution / one solution / infinitely many, via **pivots**.
- Week 3** specialises to square systems: n pivots $\iff$ invertible $\iff$ unique solution for every right-hand side. Then it looks at the *shape* of solution sets: subspaces, span, and “inhomogeneous = particular + homogeneous”.
- Week 4** asks: how many vectors do you *really* need to span a subspace? Answer: a basis; its size is the dimension. Finding dependencies is again Gauss–Jordan on $A\lambda = 0$.

Each file ends with the “slide questions” from the lectures, with worked answers.